# What momentum says about impact



**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect stopping impulse to average force without claiming a complete damage forecast.

**SUCCESS CRITERIA - I can...**

1. Compute momentum change during a stop.
2. Compare average force for different stopping times.
3. Name information beyond momentum needed for damage.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “impact” and “momentum” to explain your model or result. Sentence starters are optional.

**impact**: a collision or strike between objects

**momentum**: mass times velocity, including direction

**interval**: the time or range between two stated values

## 01 Warm-up: bring back an earlier idea

Two 3 kg carts at +5 and -3 m/s stick.
**Find** final velocity and use kinetic energy to explain why this is not elastic.

**Word help:** kinetic energy = energy associated with motion

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $p = mv$; $|F_{avg}| = |\Delta p|/\Delta t$.
Stopping force depends on stopping time as well as momentum. These quantities alone do not specify damage.

**Worked reasoning:** Two incoming objects have equal momentum 12 kg·m/s but masses 2 and 4 kg, so their speeds are 6 and 3 m/s. Momentum alone does not identify every possible consequence of stopping them.

**Your turn: Draw** equal momentum arrows and unequal velocity arrows.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .



2 kg; 6 m/s

4 kg; 3 m/s

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

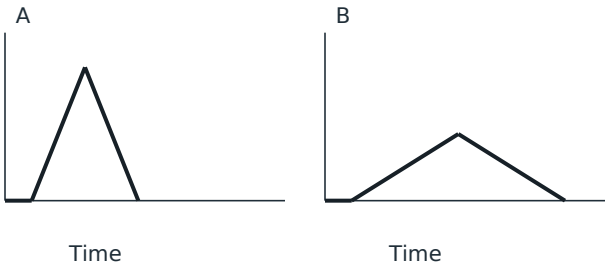
## 03 Read and interpret the model

Two equal-momentum objects stop in 0.1 s and 1 s. Which has greater average force magnitude, and by what factor?

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** Both … . They differ in … because … .

Force (same scales)




**Marzano check - circle:** 1 2 3

## 04 Apply the physics

A hypothetical object of mass $10^6$ kg at 200 m/s stops in 0.50 s.
**Find** impulse and average force magnitude. Then compare a 5 s stop.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** Both … . They differ in … because … .

| Object / trial | Mass | Velocity | Momentum |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

Why is a single stopping-force number not a prediction of buildings damaged by an asteroid?

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** 1 2 3

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Continue down each column.

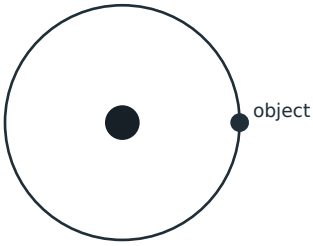
Sentence starters are optional. You may plan in any language.

## 06 Review: connect an earlier idea

An ideal circular orbit has GM=576 m³/s² and r=16 m.
**Find** speed, inward acceleration and period; explain how velocity changes.
**Label** velocity and inward acceleration on the circular model.

**Word help:** model = a drawing, explanation or equation for a physical situation

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .




**Marzano check - circle:** 1 2 3

## 07 Exit check

Objects A and B have equal momentum. A has twice B's mass.
**Compare** speeds and kinetic energies, showing why equal momentum need not mean equal energy.

**Sentence starter:** Both ... . They differ in ... because ... .

| Object / trial | Mass | Velocity | Momentum |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A 0.5 kg ball arrives at +10 m/s and either stops or rebounds at -6 m/s in 0.04 s.
**Compare** impulse and average force.

**Sentence starter:** Both ... . They differ in ... because ... .

| Object / trial | Mass | Velocity | Momentum |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3